

Description of files in this folder

1. generating_counts_table.md
 - a. File describing how the counts table (peaks x samples) was generated from BAM files.
 - i. The BAM files were produced from the raw FASTQ files available on SRA, as described in Supplementary Note 2.
 - b. There are three main steps here, as described [here](#) (PMID 33606259)
 - i. Peak calling and IDR to determine confident peaks per sample
 - ii. Merging peaks across samples
 - iii. FeatureCounts to generate the counts table
 - c. Please note, the markdown mentions/describes the following files that are also contained in this folder:
 - i. peak_merge_atac_env.yml
 - ii. for_merging_peaks subfolder
 - iii. merged_peaks subfolder
 - iv. iterative_overlap_metadata.txt
 - v. All_Samples.counts
2. deseq2.Rmd
 - a. Input:
 - i. The All_Samples.counts file generated in the previous step and contained in this folder
 - b. Outputs:
 - i. Volcano plot used for a figure panel
 - ii. DESeq2 results (deseq_results.tsv) contained in this folder
 1. Also deseq_results_with_brain3.tsv (results that include the sample with low TSS enrichment score that was excluded from the analyses in the paper, see Supplementary Note 2)
 - iii. Files needed for downstream analyses, see below
 1. sig_peak_file_for_homer.txt
 2. up_peaks_input_for_cov_heatmaps.bed
 3. down_peaks_input_for_cov_heatmaps.bed
3. homer_command.sh
 - a. Command used to run HOMER to annotate the significantly differentially accessible peaks
 - i. Requires the sig_peak_file_for_homer.txt (generated above)
4. differential_peak_annotation_analysis.Rmd
 - a. Inputs
 - i. Outputs from HOMER, contained within this folder
 1. annotation_stats.txt

- 2. annotated_sig_peak_file.txt
 - ii. deseq_results.tsv (generated above)
 - iii. DESeq2 results from the mouse brain RNA-seq experiment for GDCA
 - 1. Contained within the “Mouse brain RNAseq” folder of this repo
- b. Outputs
 - i. A pie chart showing the percentage of differentially accessible peaks annotated to each genomic category, such as introns, exons, and promoters, for use in a figure.
 - ii. A bar chart showing whether each genomic category is enriched or depleted among differentially accessible peaks relative to the genomic background, for use in a figure.
 - iii. A text file - for use in a supplementary table - containing HOMER and rGREAT annotations for each differentially accessible peak. The file also indicates whether genes linked to each peak by rGREAT showed differential expression in the RNA-seq analysis.
- 5. generating_cov_matrices.md
 - a. Markdown file describing the process that was used to generate the coverage matrices that were input into cov_heatmap.Rmd (below)
- 6. cov_heatmap.Rmd
 - a. Inputs:
 - i. The four coverage matrices generated in the previous step
 - b. Outputs:
 - i. Coverage heatmaps in saline and GDCA samples for peaks identified as differentially accessible between saline- and GDCA-treated brains.
 - 1. Also outputted the underlying data frames (derived from the coverage matrices)
- 7. chromvar.Rmd
 - a. Inputs
 - i. All_Samples.counts generated above
 - b. Key outputs – all contained within chromvar_results subfolder
 - i. A text file (TF_variability) showing variability scores across tested samples, per TF motif
 - 1. Incorporated into a supplementary table
 - ii. Heatmap showing correlations between samples based on TF motif accessibility
 - 1. Used for a figure

- iii. Heatmap of chromVAR deviation score (proxy for accessibility) for the 25 TF motifs with the greatest variability in accessibility across samples
 - 1. Used for a figure
- iv. Plot showing the variability estimates for the 10 TF motifs with the greatest variability in accessibility across samples
 - 1. Used for a figure